

Week 6

6.1 Second Derivative Test

Suppose $f : \Omega \subseteq \mathbb{R}^n \rightarrow \mathbb{R}$ is a C^2 function, and $\vec{a} \in \Omega$ is a critical point of f . Then $\nabla f(\vec{a}) = 0$. For $\vec{x}$ near $\vec{a}$, we have that $f(\vec{x}) \approx P_2(\vec{x})$. That is,

$$\begin{aligned} f(\vec{x}) - f(\vec{a}) &\approx P_2(\vec{x}) - f(\vec{a}) \\ &= \underbrace{\nabla f(\vec{a})}_{=0}(\vec{x} - \vec{a}) + \frac{1}{2}(\vec{x} - \vec{a})^T Hf(\vec{a})(\vec{x} - \vec{a}) \\ &= \frac{1}{2}(\vec{x} - \vec{a})^T Hf(\vec{a})(\vec{x} - \vec{a}). \end{aligned}$$

Therefore, the Hessian matrix can potentially determine whether $\vec{a}$ is a local extremum.

For $n = 1$, $\frac{1}{2}(x - a)^T Hf(a)(x - a) = \frac{1}{2}f''(a)(x - a)^2$, and we see that

$$\begin{cases} f''(a) > 0 & \implies \text{local min at } a, \\ f''(a) < 0 & \implies \text{local max at } a. \end{cases}$$

For $n = 2$, letting $\vec{a} = (x_0, y_0)$, we have

$$\frac{1}{2}(\vec{x} - \vec{a})^T Hf(\vec{a})(\vec{x} - \vec{a}) = \frac{1}{2}(x - x_0, y - y_0) \begin{pmatrix} f_{xx}(x_0, y_0) & f_{xy}(x_0, y_0) \\ f_{yx}(x_0, y_0) & f_{yy}(x_0, y_0) \end{pmatrix} \begin{pmatrix} x - x_0 \\ y - y_0 \end{pmatrix}.$$

To understand the nature of such critical points, we study quadratic forms (in this case of two variables)

$$q(x, y) = (x, y) \begin{pmatrix} A & B \\ B & C \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = Ax^2 + 2Bxy + Cy^2.$$

Q: Does $q(x, y)$ have a sign?

Example 123. If $q(x, y) = 2xy$, we could rewrite this as

$$q(x, y) = \frac{1}{2}(x + y)^2 - \frac{1}{2}(x - y)^2.$$

What is the sign of $q(x, y)$ away from the origin $(0, 0)$?

- Along $x + y = 0$, $q(x, -x) = -2x^2 < 0$.

- Along $x - y = 0$, $q(x, x) = 2x^2 > 0$.

Therefore, q is indefinite. Clearly, the origin is a critical point of $q(x, y)$, but neither a local minimum or a local maximum. We call such a point a saddle point.

Example 124. If $q(x, y) = 17x^2 - 12xy + 8y^2$, we could again complete the square

$$q(x, y) = 17\left(x - \frac{6}{17}y\right)^2 + \frac{100}{17}y^2. \quad (6.18)$$

In this form, it is then clear that $q(x, y) > 0 = q(0, 0)$ for any $(x, y) \neq (0, 0)$, and the critical point at the origin is a local (and global) minimum of q .

Remark. An expression such as (6.18) is known as a diagonalisation of q , and is not unique. For example, we could also write our quadratic form q as

$$q(x, y) = 5\left(\frac{x + 2y}{\sqrt{5}}\right)^2 + 20\left(\frac{2x - y}{\sqrt{5}}\right)^2.$$

Different diagonalisations are related to one another by means of an orthogonal change of coordinates.

Example 125. Consider the quadratic form $q(x, y, z) = xy + yz + xz$ of three variables. By completing the square twice, we rewrite q in the form

$$\begin{aligned} q(x, y, z) &= xy + yz + xz \\ &= \frac{1}{4}(x + y)^2 - \frac{1}{4}(x - y)^2 + z(x + y) \\ &= \frac{1}{4}(x + y + 2z)^2 - \frac{1}{4}(x - y)^2 - z^2. \end{aligned}$$

- Along the line $x - y = z = 0$, $q(x, x, 0) = x^2 > 0$.
- On the plane $x + y + 2z = 0$, $q(x, y, -\frac{1}{2}(x + y)) = -\frac{1}{4}(x - y)^2 - \frac{1}{4}(x + y)^2 < 0$.

Therefore, the critical point $(0, 0, 0)$ is a saddle point.

Definition 6.26. Let A be an $n \times n$ symmetric matrix. Then A is said to be

1. positive definite if $x^T A x > 0$ for any $x \in \mathbb{R}^n \setminus \{0\}$;
2. negative definite if $x^T A x < 0$ for any $x \in \mathbb{R}^n \setminus \{0\}$;
3. indefinite if $x^T A x > 0$ and $y^T A y < 0$, for some $x, y \in \mathbb{R}^n$.

Remark. Not every symmetric $n \times n$ matrix fits into one of these three cases.

Example 126. The matrix $A = \begin{pmatrix} 1 & 0 \\ 0 & 4 \end{pmatrix}$ is positive definite, $B = \begin{pmatrix} -1 & 0 \\ 0 & -4 \end{pmatrix}$ is negative definite, $C = \begin{pmatrix} -1 & 0 \\ 0 & 4 \end{pmatrix}$ is indefinite, and $D = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$ is neither positive definite, negative definite, or indefinite.

Contents

The following theorem, known as the Second Derivative Test, determine the nature of a critical point via the definiteness of the Hessian matrix at that point.

Theorem 127 (Second Derivative Test). *If $\Omega \subseteq \mathbb{R}^n$ is open, $f : \Omega \rightarrow \mathbb{R}$ is a C^2 function, and $\vec{a} \in \Omega$ is a critical point of f . Then $Hf(a)$ is*

(i) *positive definite $\implies \vec{a}$ is a local minimum;*

(ii) *negative definite $\implies \vec{a}$ is a local maximum;*

(iii) *indefinite $\implies \vec{a}$ is a saddle point.*

Proof. Without loss of generality, we may assume that $\vec{a} = 0$. By Taylor's theorem, since $\nabla f(0) = 0$, we have that

$$f(\vec{x}) - f(0) = \frac{1}{2} \vec{x}^T Hf(0) \vec{x} + \varepsilon(\vec{x}), \quad \forall \vec{x} \in \Omega,$$

with $\lim_{\vec{x} \rightarrow 0} \frac{\varepsilon(\vec{x})}{\|\vec{x}\|^2} = 0$. Next, consider the map $F : \partial B_1(0) \subseteq \mathbb{R}^n \rightarrow \mathbb{R}$, defined by

$$F(\vec{x}) := \frac{1}{2} \vec{x}^T Hf(0) \vec{x}, \quad \forall \text{ unit vectors } \vec{x} \in \partial B_1(0).$$

If $Hf(0)$ is positive definite, then $F > 0$. Moreover, since $\partial B_1(0)$ is compact and F is continuous (F is a polynomial of degree 2), by the EVT this map has a global minimum. In particular, there exists a unit vector $\vec{x}_0 \in \mathbb{R}^n$ such that

$$F(\vec{x}) \geq F(\vec{x}_0) > 0, \quad \forall \vec{x} \in \partial B_1(0).$$

Setting $\epsilon_0 := F(\vec{x}_0) > 0$, we find that for any non-zero vector $\vec{x} \in \mathbb{R}^n$

$$\frac{1}{2} \vec{x}^T Hf(0) \vec{x} = \frac{1}{2} \left(\frac{\vec{x}}{\|\vec{x}\|} \right)^T Hf(0) \left(\frac{\vec{x}}{\|\vec{x}\|} \right) \|\vec{x}\|^2 \geq \epsilon_0 \|\vec{x}\|^2.$$

Plugging this choice of ϵ_0 into the definition of $\lim_{\vec{x} \rightarrow 0} \frac{\varepsilon(\vec{x})}{\|\vec{x}\|^2} = 0$, we find $\delta > 0$ such that, if $\vec{x} \in \Omega$ and $0 < \|\vec{x}\| < \delta$, then $\frac{|\varepsilon(\vec{x})|}{\|\vec{x}\|^2} < \epsilon_0$. In particular, at any point $\vec{x} \in \Omega$ with $0 < \|\vec{x}\| < \delta$,

$$\begin{aligned} f(\vec{x}) - f(0) &= \frac{1}{2} \vec{x}^T Hf(0) \vec{x} + \varepsilon(\vec{x}) \\ &\geq \left| \frac{1}{2} \vec{x}^T Hf(0) \vec{x} \right| - |\varepsilon(\vec{x})| \\ &\geq \epsilon_0 \|\vec{x}\|^2 - |\varepsilon(\vec{x})| \\ &> \epsilon_0 \|\vec{x}\|^2 - \epsilon_0 \|\vec{x}\|^2 = 0, \end{aligned}$$

and hence 0 is a local minimum of f . The same argument applies if $Hf(0)$ is negative definite. Finally, in the case that $Hf(0)$ is indefinite, we find unit vectors $\vec{x}, \vec{y} \in \mathbb{R}^n$ such that

$$\alpha := \frac{1}{2} \vec{x}^T Hf(0) \vec{x} > 0, \quad \beta := \frac{1}{2} \vec{y}^T Hf(0) \vec{y} < 0.$$

Consider the sequence of vectors $\vec{x}_j := \frac{\vec{x}}{j} \in \mathbb{R}^n$. Note that $\vec{x}_j \rightarrow 0$ as $j \rightarrow \infty$. Moreover,

$$\begin{aligned} \lim_{j \rightarrow \infty} [j^2(f(\vec{x}_j) - f(0))] &= \lim_{j \rightarrow \infty} j^2 \left[\frac{1}{2} \vec{x}_j^T Hf(0) \vec{x}_j + \varepsilon(\vec{x}_j) \right] \\ &= \frac{1}{2} \vec{x}^T Hf(0) \vec{x} + \lim_{j \rightarrow \infty} \frac{\varepsilon(\vec{x}_j)}{\|\vec{x}_j\|^2} \\ &= \alpha + 0 = \alpha > 0. \end{aligned}$$

Therefore, for j sufficiently large, $f(\vec{x}_j) > f(0)$. Repeating the argument for the sequence $\vec{y}_j = \frac{\vec{y}}{j}$, we find that for j sufficiently large, $f(\vec{y}_j) < f(0)$. Thus we conclude that 0 is a saddle point. $\square$

6.1.1 The $n = 2$ Case

In practise, how do we determine if the Hessian matrix is definite? In the 2-dimensional case, we have the following easy test:

Lemma 128. Let $M = \begin{pmatrix} A & B \\ B & C \end{pmatrix}$ be a 2×2 matrix. Then

- M is positive definite $\iff A > 0$, and $\det(M) > 0$;
- M is negative definite $\iff A < 0$, and $\det(M) > 0$;
- M is indefinite $\iff \det(M) < 0$.

Proof. Let $q(x, y) = Ax^2 + 2Bxy + Cy^2$.

(i) We begin with the case that $A \neq 0$:

$$Aq(x, y) = A^2x^2 + 2ABxy + ACy^2 = (Ax + By)^2 + \underbrace{(AC - B^2)}_{\det M} y^2.$$

From this we conclude that $Aq > 0$ if $\det(M) > 0$, and Aq changes sign if $\det(M) < 0$. Dividing by A gives us that

- $q > 0 \iff A > 0, \det(M) > 0$;
- $q < 0 \iff A < 0, \det(M) > 0$;
- q changes sign $\iff \det(M) < 0$.

(ii) The case $A = 0$:

$$q(x, y) = y(2Bx + Cy),$$

is clearly neither positive definite nor negative definite. q is indefinite $\iff B \neq 0 \iff \det(M) = -B^2 < 0$. $\square$

We use this Lemma to update the Second Derivative Test in the case $n = 2$.

Theorem 129 (Second Derivative Test, $n = 2$). If $\Omega \subseteq \mathbb{R}^2$ is open, $f : \Omega \rightarrow \mathbb{R}$ is a C^2 function, and $\vec{a} \in \Omega$ is a critical point of f . Then

- (i) $f_{xx}f_{yy} - f_{xy}^2 > 0, f_{xx} > 0$ at $\vec{a} \implies \vec{a}$ is a local minimum;
- (ii) $f_{xx}f_{yy} - f_{xy}^2 > 0, f_{xx} < 0$ at $\vec{a} \implies \vec{a}$ is a local maximum;
- (iii) $f_{xx}f_{yy} - f_{xy}^2 < 0$ at $\vec{a} \implies \vec{a}$ is a saddle point;
- (iv) $f_{xx}f_{yy} - f_{xy}^2 = 0$ at $\vec{a} \implies$ inconclusive.

Remark. Case (iv) could be a local max/min or a saddle point (see Example 132).

Example 130. Consider the function $f : \mathbb{R}^2 \rightarrow \mathbb{R}$, $f(x, y) = 3x^2 - 10xy + 3y^2 + 2x + 2y + 3$. This is a smooth function with

$$\nabla f(x, y) = (6x - 10y + 2, -10x + 6y + 2),$$

and so the only critical point of f is at $(x, y) = (\frac{1}{2}, \frac{1}{2})$. The Hessian matrix of f is

$$Hf(x, y) = \begin{pmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{pmatrix} = \begin{pmatrix} 6 & -10 \\ -10 & 6 \end{pmatrix}.$$

Note that (at the critical point)

$$f_{xx}f_{yy} - f_{xy}^2 = 6^2 - 10^2 = -64 < 0,$$

and by the Second Derivative Test, $(\frac{1}{2}, \frac{1}{2})$ is a saddle point.

Example 131. Consider the function $f : \mathbb{R}^2 \rightarrow \mathbb{R}$, $f(x, y) = 3x - x^3 - 3xy^2$. This is a smooth function with

$$\nabla f(x, y) = (3 - 3x^2 - 3y^2, -6xy),$$

and so the critical points of f are when

$$xy = 0, x^2 + y^2 = 1 \iff (x, y) = (0, \pm 1), (\pm 1, 0).$$

The Hessian matrix of f is

$$Hf(x, y) = \begin{pmatrix} -6x & -6y \\ -6y & -6x \end{pmatrix}.$$

Therefore, at each critical point we deduce that:

- $Hf(1, 0) = \begin{pmatrix} -6 & 0 \\ 0 & -6 \end{pmatrix}$, with $\det(Hf(1, 0)) = 36 > 0$ and $f_{xx} = -6 < 0$. So $(1, 0)$ is a local maximum.
- $Hf(-1, 0) = \begin{pmatrix} 6 & 0 \\ 0 & 6 \end{pmatrix}$, with $\det(Hf(-1, 0)) = 36 > 0$ and $f_{xx} = 6 > 0$. So $(-1, 0)$ is a local minimum.

- $Hf(0, 1) = \begin{pmatrix} 0 & -6 \\ -6 & 0 \end{pmatrix}$, with $\det(Hf(0, 1)) = -36 < 0$. So $(0, 1)$ is a saddle point.
- $Hf(0, -1) = \begin{pmatrix} 0 & 6 \\ 6 & 0 \end{pmatrix}$, with $\det(Hf(0, -1)) = -36 < 0$. So $(0, -1)$ is a saddle point.

Example 132. Consider the three functions

$$f(x, y) = x^2 + y^4, \quad g(x, y) = x^2 - y^4, \quad h(x, y) = -x^2 - y^4.$$

Since

$$\nabla f(x, y) = (2x, 4y^3), \quad \nabla g(x, y) = (2x, -4y^3), \quad \nabla h(x, y) = (-2x, -4y^3),$$

all three functions have a single critical point at $(0, 0)$. Since

$$Hf(x, y) = \begin{pmatrix} 2 & 0 \\ 0 & 12y^2 \end{pmatrix}, \quad Hg(x, y) = \begin{pmatrix} 2 & 0 \\ 0 & -12y^2 \end{pmatrix}, \quad Hh(x, y) = \begin{pmatrix} -2 & 0 \\ 0 & -12y^2 \end{pmatrix},$$

and

$$Hf(0, 0) = \begin{pmatrix} 2 & 0 \\ 0 & 0 \end{pmatrix}, \quad Hg(0, 0) = \begin{pmatrix} 2 & 0 \\ 0 & 0 \end{pmatrix}, \quad Hh(0, 0) = \begin{pmatrix} -2 & 0 \\ 0 & 0 \end{pmatrix},$$

all three Hessian matrices have zero determinant at the origin, and the Second Derivative Test is inconclusive. However at $(0, 0)$, we see that f has a local minimum, g has a saddle point, and h has a local maximum.

6.1.2 The General Case (non-examinable)

Let us return to the general case $f : \Omega \subseteq \mathbb{R}^n \rightarrow \mathbb{R}$ is a C^2 function, $\vec{a} \in \Omega$ and $\vec{a}$ is a critical point of f . Then

$$Hf(a) = \begin{pmatrix} f_{11} & \cdots & f_{1n} \\ \vdots & \ddots & \vdots \\ f_{n1} & \cdots & f_{nn} \end{pmatrix}.$$

Since f is C^2 , $Hf(\vec{a})$ is symmetric, and thus by a theorem from Linear Algebra, there exists an orthogonal matrix P such that

$$P^T Hf(\vec{a}) P = \begin{pmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{pmatrix},$$

is a diagonal matrix, where $\lambda_1, \dots, \lambda_n \in \mathbb{R}$ are the eigenvalues of $Hf(\vec{a})$.

(Recall, a matrix P is called orthogonal if $P^T P = I_n$.)

Note that, the diagonal matrix $P^T Hf(\vec{a}) P$ is

- positive definite $\iff \lambda_i > 0$ for any $i \in \{1, \dots, n\}$;

- negative definite $\iff \lambda_i < 0$ for any $i \in \{1, \dots, n\}$;
- indefinite $\iff \lambda_i < 0 < \lambda_j$ for some $i, j \in \{1, \dots, n\}$.

Since P is invertible ($P^{-1} = P^T$), we see that the previous criteria also apply to the matrix $Hf(\vec{a})$.

Alternatively, we can also check the definiteness of the Hessian matrix by looking at submatrices. That is, let H_k denote the $k \times k$ submatrix

$$H_k = \begin{pmatrix} f_{11} & \cdots & f_{1k} \\ \vdots & \ddots & \vdots \\ f_{k1} & \cdots & f_{kk} \end{pmatrix}, \quad \text{for } k \in \{1, \dots, n\}.$$

Then, $Hf(a)$ is

- positive definite $\iff \det(H_k) > 0$, for $k \in \{1, \dots, n\}$;
- negative definite $\iff \text{sign}(\det(H_k)) = (-1)^k$, for $k \in \{1, \dots, n\}$.

For $n = 2$,

$$\det(H_1) = f_{xx}, \quad \det(H_2) = f_{xx}f_{yy} - f_{xy}^2,$$

and we recover the $n = 2$ version of the Second Derivative Test stated earlier.

6.2 Lagrange Multipliers

To motivate the next section, we consider the following question:

what if we want extrema subject to constraints?

Theorem 133 (Lagrange Multiplier Theorem). *Let $f, g : \Omega \subseteq \mathbb{R}^n \rightarrow \mathbb{R}$ be C^1 functions, $c \in \mathbb{R}$, and*

$$S = g^{-1}(c) = \{x \in \Omega : g(x) = c\}.$$

If $\vec{a}$ is a local extremum of f on S , and $\nabla g(\vec{a}) \neq 0$, then $\nabla f(\vec{a}) = \lambda \nabla g(\vec{a})$ for some $\lambda \in \mathbb{R}$.

Remark. *Let $F : \Omega \times \mathbb{R} \rightarrow \mathbb{R}$ denote the function $F(x, \lambda) = f(x) - \lambda(g(x) - c)$. Then*

$$\nabla F(x, \lambda) = (\nabla f(x) - \lambda \nabla g(x), c - g(x)).$$

Therefore, critical points of f under the constraint $g \equiv c$ are equivalent to unconstrained critical points of F .

The set of equations:

$$\begin{aligned} \nabla f(\vec{a}) &= \lambda \nabla g(\vec{a}) \\ g(\vec{a}) &= c \end{aligned}$$

are called the *Lagrange multiplier equations* (λ is the Lagrange multiplier). Solving them gives us candidates for extrema of f on $\{g = c\}$ (along with perhaps the points where $\nabla g = 0$).

Example 134. Suppose we want to find the point on the parabola $4y = x^2$ closest to the point $(1, 2)$.

We may rephrase this by introducing the squared distance function $f(x, y) = (x - 1)^2 + (y - 2)^2$, and the constraining function $g(x, y) = x^2 - 4y$. Then our question becomes find the minimum of f on the level set $g = 0$.

Since f, g are polynomials on $\mathbb{R}^2$ they are certainly C^1 on $\mathbb{R}^2$ with

$$\nabla f = (2(x - 1), 2(y - 2)), \quad \nabla g = (2x, -4) \neq 0.$$

Thus, if (x, y) is a local extremum of f on $g = 0$, then

$$\begin{cases} \nabla f(x, y) = \lambda \nabla g(x, y) & : \text{ for some } \lambda \in \mathbb{R} \\ g(x, y) = 0, \end{cases}$$

which is equivalent to the system of equations

$$2(x - 1) = 2\lambda x, \quad 2(y - 2) = -4\lambda, \quad x^2 = 4y.$$

Solving this system, we find a unique solution $(x, y) = (2, 1)$.

Geometrically, f must have a minimum on $g = 0$. By our reasoning, $(2, 1)$ is the only possible candidate, and thus f has a minimum at $(2, 1)$ on the level set $g \equiv 0$. Since $f(2, 1) = 2$, in the original language of the question, the point $(2, 1)$ is the point lying on the parabola $4y = x^2$ lying closest to $(1, 2)$, with a distance of $\sqrt{2}$ between the points.

Idea of Proof. Suppose f has an extremum on S at a . Let $\gamma : (\epsilon, \epsilon) \rightarrow S$ be a curve on S such that $\gamma(0) = a$. Since $f \circ \gamma$ has an extremum at 0,

$$0 = (f \circ \gamma)'(0) = \nabla f(a) \cdot \gamma'(0),$$

and so $\nabla f(a) \perp S$ at a . Since $\nabla g(a)$ is also normal to S at a , we find that $\nabla f(a)$ and $\nabla g(a)$ are parallel. $\square$

Example 135. Find the point on the parabola $x^2 = 4y$ closest to the point $(2, 5)$. As before, we set up the problem in the following way:

$$\text{Minimise } f(x, y) = (x - 2)^2 + (y - 5)^2 \text{ with the constraint } g(x, y) = x^2 - 4y = 0.$$

As in the previous example, f, g are C^1 functions with

$$\nabla f = (2(x - 2), 2(y - 5)), \quad \nabla g = (2x, -4) \neq 0.$$

Thus, if (x, y) is a local extremum of f on $g = 0$, then

$$\begin{cases} \nabla f(x, y) = \lambda \nabla g(x, y) & : \text{ for some } \lambda \in \mathbb{R} \\ g(x, y) = 0, \end{cases}$$

which is equivalent to the system of equations

$$2(x - 2) = 2\lambda x, \quad 2(y - 5) = -4\lambda, \quad x^2 = 4y.$$

Solving this system, we now find two solutions $(x, y) = (-2, 1), (4, 4)$.

Again, geometrically, f must have a minimum on $g = 0$. Since $f(-2, 1) = 32$ and $f(4, 4) = 5$, f has a minimum at $(4, 4)$ on the level set $g \equiv 0$, and the point $(4, 4)$ is the point lying on the parabola $4y = x^2$ lying closest to $(2, 5)$.

Example 136. Maximise the function $f(x, y) = xy^2$ on the ellipse $g(x, y) = x^2 + 4y^2 = 4$.

Since f is continuous and the level set $\{g = 4\}$ is compact, by the EVT f has a global maximum on the ellipse. Since f, g are C^1 functions with

$$\nabla f = (y^2, 2xy), \quad \nabla g = (2x, 8y),$$

and $\nabla g \neq 0$ on the ellipse, we may use the theory of Lagrange Multipliers to conclude that at such a maximum

$$\begin{cases} \nabla f(x, y) = \lambda \nabla g(x, y) & : \text{ for some } \lambda \in \mathbb{R} \\ g(x, y) = 4. \end{cases}$$

This is equivalent to the system of equations

$$y^2 = 2\lambda x, \quad 2xy = 8\lambda y, \quad x^2 + 4y^2 = 4.$$

If $y = 0$, then $x = \pm 2$. Otherwise $y \neq 0$, and from our equations we deduce that

$$\frac{2xy}{y} = \frac{8\lambda y}{2\lambda x} \implies x^2 = 2y^2,$$

and hence $(x, y) = (\pm\sqrt{\frac{4}{3}}, \pm\sqrt{\frac{2}{3}})$. Comparing our function at these six points, we find that

$$f(\pm 2, 0) = 0, \quad \underbrace{f\left(\sqrt{\frac{4}{3}}, \pm\sqrt{\frac{2}{3}}\right) = \frac{4}{3\sqrt{3}}, \quad f\left(-\sqrt{\frac{4}{3}}, \pm\sqrt{\frac{2}{3}}\right) = -\frac{4}{3\sqrt{3}}}_{\text{Maximum}}.$$

For problems in finding unconstrained extremal points for functions $f : A \rightarrow \mathbb{R}$, the theory of Lagrange Multipliers can be utilised to determine the extremal points of f on the boundary ∂A .

Example 137. In Example 114, we found that $f(x, y) = x^2 + 2y^2 - x + 3$ has the single critical point $(\frac{1}{2}, 0) \in \text{Int}(A) = B_1(0)$, with $f(\frac{1}{2}, 0) = \frac{11}{4}$. Instead of parameterising the boundary ∂A , we instead define $g(x, y) = x^2 + y^2$ and note that $\partial A = \{g = 1\}$. Since $\nabla g = (2x, 2y) \neq 0$ on ∂A , we may apply the theory of Lagrange Multipliers to deduce that at an extrema of f restricted to ∂A , we must satisfy the system of equations

$$\begin{cases} \nabla f(x, y) = \lambda \nabla g(x, y) & : \text{ for some } \lambda \in \mathbb{R} \\ g(x, y) = 1, \end{cases}$$

which is equivalent to

$$2x - 1 = 2\lambda x, \quad 4y = 2\lambda y, \quad x^2 + y^2 = 1.$$

From the second equation, we deduce that either $y = 0$, in which case $x = \pm 1$, or $\lambda = 2$, in which case $(x, y) = (-\frac{1}{2}, \pm \frac{\sqrt{3}}{2})$. Comparing f at these points

$$\underbrace{f(\frac{1}{2}, 0) = \frac{11}{4}}_{\text{Minimum}}, \quad f(1, 0) = 3, \quad f(-1, 0) = 5, \quad , \underbrace{f\left(-\frac{1}{2}, \pm \frac{\sqrt{3}}{2}\right) = \frac{21}{4}}_{\text{Maximum}}.$$

The following theorem generalises to the case of multiple constraints.

Theorem 138. Let $f, g_1, \dots, g_k : \Omega \subseteq \mathbb{R}^n \rightarrow \mathbb{R}$ be C^1 functions, $c \in \mathbb{R}^k$, and

$$S = \{x \in \Omega : g_i(x) = c_i, i \in \{1, \dots, k\}\}.$$

If $\vec{a}$ is a local extremum of f on S , and $\nabla g_1(\vec{a}), \dots, \nabla g_k(\vec{a})$ are linearly independent, then

$$\nabla f(\vec{a}) = \sum_{i=1}^k \lambda_i \nabla g_i(\vec{a}), \quad \text{for some } \lambda \in \mathbb{R}^k.$$

Idea of proof. Just like in the case of a single constraint, $\nabla f(\vec{a}) \perp S$ at a . That is, $\nabla f(\vec{a})$ is perpendicular to the $(n - k)$ -dimensional tangent plane to S at $\vec{a}$. Therefore, $\nabla f(\vec{a})$ must belong to the span of the normal vectors $\nabla g_1(\vec{a}), \dots, \nabla g_k(\vec{a})$. $\square$

Example 139. Maximise the function $f(x, y, z) = x^2 + 2y^2 - z^2$ on the line L given by the two equations $2x - y = 0$, $y + z = 0$, assuming that we know f does admit a maximum on L .

Defining our constraints $g_1(x, y, z) = 2x - y$, $g_2(x, y, z) = y + z$, we find that

$$\nabla f = (2x, 2, -2z), \quad \nabla g_1 = (2, -1, 0), \quad \nabla g_2 = (0, 1, 1),$$

with $\nabla g_1(x, y, z), \nabla g_2(x, y, z)$ linearly independent at every point $(x, y, z) \in \mathbb{R}^3$. Therefore, we may use the theory of Lagrange Multipliers to deduce that at the maximum of f on L , we satisfy the system of equations

$$\begin{cases} \nabla f(x, y, z) = \lambda_1 \nabla g_1(x, y, z) + \lambda_2 \nabla g_2(x, y, z) & : \text{ for some } \lambda_1, \lambda_2 \in \mathbb{R} \\ g_1(x, y, z) = 0, \\ g_2(x, y, z) = 0, \end{cases}$$

which is equivalent to the system of equations

$$2x = 2\lambda_1, \quad 2 = -\lambda_1 + \lambda_2, \quad -2z = \lambda_2, \quad 2x - y = 0, \quad y + z = 0.$$

This system has the unique solution $(x, y, z) = (\frac{2}{3}, \frac{4}{3}, -\frac{4}{3})$. Since a maximum exists, it must occur at this point.

Example 140. Find the distance between the curve $xy = 1$ and the line $x + 4y = \frac{15}{8}$ in $\mathbb{R}^2$.

Given a point (x, y) on the curve and a point (u, v) on the line, the distance squared between them is given by the function $f(x, y, u, v) = (x - u)^2 + (y - v)^2$. The point (x, y) lying on the curve can be expressed as the constraint $g_1(x, y, u, v) = xy = 1$, and similarly $g_2(x, y, u, v) = u + 4v + \frac{15}{8}$ is equivalent to (u, v) lying on the line. Therefore, we need to minimise f under the constraints $g_1 = 1, g_2 = \frac{15}{8}$. Since

$$\begin{aligned}\nabla f &= (2(x - u), 2(y - v), -2(x - u), -2(y - v)) \\ \nabla g_1 &= (y, x, 0, 0) \\ \nabla g_2 &= (0, 0, 1, 4),\end{aligned}$$

$\nabla g_1, \nabla g_2$ are linearly independent away from the origin $(0, 0)$. But $xy = 1$, so these vectors are linearly independent when our constraints are satisfied, and hence by the theory of Lagrange Multipliers, we know that at a minimum, we solve the system of equations

$$\begin{cases} \nabla f(x, y, u, v) = \lambda_1 \nabla g_1(x, y, u, v) + \lambda_2 \nabla g_2(x, y, u, v) & : \text{ for some } \lambda_1, \lambda_2 \in \mathbb{R} \\ g_1(x, y, u, v) = 1, \\ g_2(x, y, u, v) = \frac{15}{8}. \end{cases}$$

This system of equations has two solutions: $(2, \frac{1}{2}, \frac{15}{8}, 0)$, and $(-2, -\frac{1}{2}, -\frac{225}{136}, \frac{15}{17})$. Since

$$\begin{aligned}f(2, \frac{1}{2}, \frac{15}{8}, 0) &= \left(\frac{1}{8}\right)^2 + \left(\frac{1}{2}\right)^2 = \frac{17}{64}, \\ f(-2, -\frac{1}{2}, -\frac{225}{136}, \frac{15}{17}) &= \underbrace{\left(2 - \frac{225}{136}\right)^2}_{\geq 0} + \underbrace{\left(\frac{15}{17} + \frac{1}{2}\right)^2}_{\geq 1} \geq 1 > \frac{17}{64},\end{aligned}$$

f attains its minimum at $(2, \frac{1}{2}, \frac{15}{8}, 0)$, and the distance between the curve and the line is

$$\sqrt{f(2, \frac{1}{2}, \frac{15}{8}, 0)} = \frac{\sqrt{17}}{8}.$$

6.3 Implicit Function Theorem

We now move away from finding extremal points, and instead ask the following:

when can we ‘solve’ a constraint?

For example, if we have an implicit relation between two variables $g(x, y) = c$, when can we find, *locally*, an explicit formula $y = f(x)$ such that $g(x, f(x)) = c$?

Example 141. Consider the level set $g(x, y) = x^2 - y^2 = 0$. Near the point:

- $(1, 1)$, we find $y = x$;

- $(-1, 1)$, we find $y = -x$;
- $(0, 0)$, y is not uniquely determined by x .

Example 142. Consider the constraint $x^2 + y^2 + z^2 = 2$ in $\mathbb{R}^3$, and consider the point $(x, y, z) = (0, 1, 1)$ satisfying the constraint. Can we locally solve z as a function of x and y near $(0, 1, 1)$? What about x as a function of y and z ?

Assuming we can solve $z = z(x, y)$ near $(0, 1, 1)$, and it is a differentiable function, by implicit differentiation we find

$$\begin{cases} 2x + 2z \frac{\partial z}{\partial x} = 0 \\ 2y + 2z \frac{\partial z}{\partial y} = 0 \end{cases},$$

which at $(0, 1, 1)$ gives $(\frac{\partial z}{\partial x}, \frac{\partial z}{\partial y}) = (0, -1)$.

However, assuming we can solve $x = x(y, z)$ near $(0, 1, 1)$, and it is a differentiable function, by implicit differentiation we find

$$\begin{cases} 2x \frac{\partial x}{\partial y} + 2y = 0 \\ 2x \frac{\partial x}{\partial z} + 2z = 0 \end{cases},$$

which at $(0, 1, 1)$ gives the contradiction $0 = 2$. Therefore, we conclude that even though it may be possible to express $x = x(y, z)$ near $(0, 1, 1)$, such a function cannot be differentiable.

Setting $g(x, y, z) = x^2 + y^2 + z^2$ so that our constraint is $g = 2$, we find that at $(0, 1, 1)$:

$$\begin{aligned} \frac{\partial g}{\partial z}(0, 1, 1) &= 2 \neq 0, \\ \frac{\partial g}{\partial x}(0, 1, 1) &= 0. \end{aligned}$$

More generally, given a constraint $F(x, y, z) = c$, if $z = z(x, y)$, then by implicit differentiation

$$\begin{cases} \frac{\partial F}{\partial x} + \frac{\partial F}{\partial z} \frac{\partial z}{\partial x} = 0 \\ \frac{\partial F}{\partial y} + \frac{\partial F}{\partial z} \frac{\partial z}{\partial y} = 0 \end{cases}. \quad (6.19)$$

If $F(\alpha, \beta, \gamma) = c$ and $\frac{\partial F}{\partial z}(\alpha, \beta, \gamma) \neq 0$, then (6.19) has a solution. Therefore $z = z(x, y)$ may exist and be differentiable with

$$\left(\frac{\partial z}{\partial x}, \frac{\partial z}{\partial y} \right) |_{(\alpha, \beta, \gamma)} = - \left(\frac{\partial F}{\partial x}(\alpha, \beta, \gamma) \right)^{-1} \left(\frac{\partial F}{\partial x}, \frac{\partial F}{\partial y} \right) |_{(\alpha, \beta, \gamma)}.$$

Example 143. Consider the curve in $\mathbb{R}^3$ given by the pair of constraints

$$\begin{cases} x^2 + y^2 + z^2 = 2, \\ x + z = 1. \end{cases}.$$

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Locally about the point $(0, 1, 1)$ is the curve really a one-dimensional curve? That is, near $(0, 1, 1)$, can we express $y = y(x)$ and $z = z(x)$?

If we assume so, then by implicit differentiation

$$\begin{cases} 2x + 2y \frac{dy}{dx} + 2z \frac{dz}{dx} = 0, \\ 1 + \frac{dz}{dx} = 0. \end{cases}$$

We can rewrite this linear system of equations in the form

$$\begin{pmatrix} 2y & 2z \\ 0 & 1 \end{pmatrix} \begin{pmatrix} \frac{dy}{dx} \\ \frac{dz}{dx} \end{pmatrix} = \begin{pmatrix} -2x \\ -1 \end{pmatrix}.$$

At the point $(0, 1, 1)$, this equation becomes

$$\begin{pmatrix} 2 & 2 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} \frac{dy}{dx} \\ \frac{dz}{dx} \end{pmatrix} = \begin{pmatrix} 0 \\ -1 \end{pmatrix},$$

which has solution

$$\begin{pmatrix} \frac{dy}{dx} \\ \frac{dz}{dx} \end{pmatrix} = \begin{pmatrix} 2 & 2 \\ 0 & 1 \end{pmatrix}^{-1} \begin{pmatrix} 0 \\ -1 \end{pmatrix} = \begin{pmatrix} \frac{1}{2} & -1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 0 \\ -1 \end{pmatrix} = \begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

More generally, given a pair of constraints $F_i(x, y, z) = c_i$ (for $i = 1, 2$), if $y = y(x)$ and $z = z(x)$, then by implicit differentiation

$$\frac{\partial F_i}{\partial x} + \frac{\partial F_i}{\partial y} \frac{dy}{dx} + \frac{\partial F_i}{\partial z} \frac{dz}{dx} = 0, \quad (\text{for } i = 1, 2),$$

which can be rewritten in the form

$$\begin{pmatrix} \frac{\partial F_1}{\partial y} & \frac{\partial F_1}{\partial z} \\ \frac{\partial F_2}{\partial y} & \frac{\partial F_2}{\partial z} \end{pmatrix} \begin{pmatrix} \frac{dy}{dx} \\ \frac{dz}{dx} \end{pmatrix} + \begin{pmatrix} \frac{\partial F_1}{\partial x} \\ \frac{\partial F_2}{\partial x} \end{pmatrix} = 0. \quad (6.20)$$

If $F_i(\alpha, \beta, \gamma) = c_i$ (for $i = 1, 2$), and the matrix $\begin{pmatrix} \frac{\partial F_1}{\partial y} & \frac{\partial F_1}{\partial z} \\ \frac{\partial F_2}{\partial y} & \frac{\partial F_2}{\partial z} \end{pmatrix} |_{(\alpha, \beta, \gamma)}$ is invertible, then (6.20) has the solution

$$\begin{pmatrix} \frac{dy}{dx} \\ \frac{dz}{dx} \end{pmatrix} |_{(\alpha, \beta, \gamma)} = - \begin{pmatrix} \frac{\partial F_1}{\partial y}(\alpha, \beta, \gamma) & \frac{\partial F_1}{\partial z}(\alpha, \beta, \gamma) \\ \frac{\partial F_2}{\partial y}(\alpha, \beta, \gamma) & \frac{\partial F_2}{\partial z}(\alpha, \beta, \gamma) \end{pmatrix}^{-1} \begin{pmatrix} \frac{\partial F_1}{\partial x} \\ \frac{\partial F_2}{\partial x} \end{pmatrix} |_{(\alpha, \beta, \gamma)}.$$

Suppose we are given $n + k$ -variables and k -equations

$$\begin{cases} F_1(x_1, \dots, x_n, y_1, \dots, y_k) = c_1, \\ \vdots \\ F_k(x_1, \dots, x_n, y_1, \dots, y_k) = c_k, \end{cases}$$

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When can we (locally) express the variables $y_1, \dots, y_k$ as functions of $x_1, \dots, x_n$?

Theorem 144 (Implicit Function Theorem). *Let $\Omega \subseteq \mathbb{R}^{n+k}$ be open and $F : \Omega \rightarrow \mathbb{R}^k$ be a C^1 -function. Denote $x = (x_1, \dots, x_n) \in \mathbb{R}^n$, $y = (y_1, \dots, y_k) \in \mathbb{R}^k$, and*

$$F(x, y) = \begin{pmatrix} F_1(x, y) \\ \vdots \\ F_k(x, y) \end{pmatrix} = \begin{pmatrix} F_1(x_1, \dots, x_n, y_1, \dots, y_k) \\ \vdots \\ F_k(x_1, \dots, x_n, y_1, \dots, y_k) \end{pmatrix}.$$

Suppose $(\vec{a}, \vec{b}) \in \Omega$ is such that $F(\vec{a}, \vec{b}) = \vec{c} \in \mathbb{R}^k$, and that the $k \times k$ matrix

$$\frac{\partial F}{\partial y}(\vec{a}, \vec{b}) = \begin{pmatrix} \frac{\partial F_1}{\partial y_1}(\vec{a}, \vec{b}) & \cdots & \frac{\partial F_1}{\partial y_k}(\vec{a}, \vec{b}) \\ \vdots & \ddots & \vdots \\ \frac{\partial F_k}{\partial y_1}(\vec{a}, \vec{b}) & \cdots & \frac{\partial F_k}{\partial y_k}(\vec{a}, \vec{b}) \end{pmatrix},$$

is invertible. Then, there exists open sets $U \subseteq \mathbb{R}^n$, $V \subseteq \mathbb{R}^k$ with $\vec{a} \in U$ and $\vec{b} \in V$, and a unique function $\varphi : U \rightarrow V$ such that $\varphi(\vec{a}) = \vec{b}$ and

$$F(x, \varphi(x)) = \vec{c}, \quad \forall x \in U.$$

Moreover, φ is a C^1 function with Jacobian matrix

$$\underbrace{\left(\frac{\partial \varphi}{\partial x} \right)}_{k \times n} = - \underbrace{\left(\frac{\partial F}{\partial y} \right)^{-1}}_{k \times k} \cdot \underbrace{\left(\frac{\partial F}{\partial x} \right)}_{k \times n}, \quad \forall \vec{x} \in U.$$

Written in full, this is the equation

$$\begin{pmatrix} \frac{\partial \varphi_1}{\partial x_1}(\vec{x}) & \cdots & \frac{\partial \varphi_1}{\partial x_n}(\vec{x}) \\ \vdots & \ddots & \vdots \\ \frac{\partial \varphi_k}{\partial x_1}(\vec{x}) & \cdots & \frac{\partial \varphi_k}{\partial x_n}(\vec{x}) \end{pmatrix} = - \begin{pmatrix} \frac{\partial F_1}{\partial y_1}(\vec{x}, \varphi(\vec{x})) & \cdots & \frac{\partial F_1}{\partial y_k}(\vec{x}, \varphi(\vec{x})) \\ \vdots & \ddots & \vdots \\ \frac{\partial F_k}{\partial y_1}(\vec{x}, \varphi(\vec{x})) & \cdots & \frac{\partial F_k}{\partial y_k}(\vec{x}, \varphi(\vec{x})) \end{pmatrix}^{-1} \begin{pmatrix} \frac{\partial F_1}{\partial x_1}(\vec{x}, \varphi(\vec{x})) & \cdots & \frac{\partial F_1}{\partial x_n}(\vec{x}, \varphi(\vec{x})) \\ \vdots & \ddots & \vdots \\ \frac{\partial F_k}{\partial x_1}(\vec{x}, \varphi(\vec{x})) & \cdots & \frac{\partial F_k}{\partial x_n}(\vec{x}, \varphi(\vec{x})) \end{pmatrix}.$$

Example 145 (The special case $k = 1$). $F : \Omega \subseteq \mathbb{R}^{n+1} \rightarrow \mathbb{R}$, with $F(\vec{x}, y) = F(x_1, \dots, x_n, y)$. That is, there is a single constraint $F(\vec{x}, y) = c \in \mathbb{R}$. Suppose $\vec{a} \in \mathbb{R}^n$ and $b \in \mathbb{R}$ such that $(\vec{a}, b) \in \Omega$ with $F(\vec{a}, b) = c$. The Implicit Function Theorem tells you that if $\frac{\partial F}{\partial y}(\vec{a}, b) \neq 0$, then we can find a differentiable function $y = y(x_1, \dots, x_n)$ locally near $\vec{x} = \vec{a}$, with $y(a) = b$, solving the constraint

$$F(x_1, \dots, x_n, y(x_1, \dots, x_n)) = c.$$

Example 146 (The special case $n = 1, k = 2$). $F : \Omega \subseteq \mathbb{R}^3 \rightarrow \mathbb{R}^2$, with $F(x, y, z) = F(x, y_1, y_2)$. Given a pair of constraints $F(x, y, z) = \vec{c} \in \mathbb{R}^2$, suppose $a \in \mathbb{R}$ and $\vec{b} \in \mathbb{R}^2$ is such that $(a, \vec{b}) \in \Omega$ with $F(a, \vec{b}) = \vec{c}$. The Implicit Function Theorem tells you that if the matrix

$$\begin{pmatrix} \frac{\partial F_1}{\partial y_1}(a, \vec{b}) & \frac{\partial F_1}{\partial y_2}(a, \vec{b}) \\ \frac{\partial F_2}{\partial y_1}(a, \vec{b}) & \frac{\partial F_2}{\partial y_2}(a, \vec{b}) \end{pmatrix},$$

is invertible, then we can find differentiable functions $y_1 = y_1(x)$, $y_2 = y_2(x)$ locally near $x = a$, with $(y_1(a), y_2(a)) = \vec{b}$, solving the constraint

$$F(x, y_1(x), y_2(x)) = \vec{c}.$$

Remark. We write $F = F(x, y)$ and solve y as a function of x , but in fact, the ordering of the variables is irrelevant; we just need to check the invertibility of the appropriate submatrix of the Jacobian matrix $DF(x, y)$.

Example 147. Consider the constraints

$$\begin{cases} xz + \sin(yz - x^2) = 8, \\ x + 4y + 3z = 18. \end{cases}$$

Near $(2, 1, 4)$, can we locally solve for two of the variables as functions of the third?

Letting $F(x, y, z) = (xz + \sin(yz - x^2), x + 4y + 3z)$, we find that

$$DF(x, y, z) = \begin{pmatrix} z - 2x \cos(yz - x^2) & z \cos(yz - x^2) & x + y \cos(yz - x^2) \\ 1 & 4 & 3 \end{pmatrix},$$

and hence

$$DF(2, 1, 4) = \begin{pmatrix} 0 & 4 & 3 \\ 1 & 4 & 3 \end{pmatrix}.$$

By looking at the invertibility of 2×2 submatrices we can apply the Implicit Function Theorem.

- $\begin{vmatrix} 0 & 4 \\ 1 & 4 \end{vmatrix} = -4 \neq 0$, and hence by the Implicit Function Theorem, x and y can be expressed as functions of z near $(2, 1, 4)$.
- $\begin{vmatrix} 0 & 3 \\ 1 & 3 \end{vmatrix} = -3 \neq 0$, and hence by the Implicit Function Theorem, x and z can be expressed as functions of y near $(2, 1, 4)$.
- $\begin{vmatrix} 4 & 3 \\ 4 & 3 \end{vmatrix} = 0$, and so no conclusion can be drawn from the Implicit Function Theorem in regards to whether y and z can be expressed as functions of x near $(2, 1, 4)$.

6.4 Inverse Function Theorem

Theorem 148 (Inverse Function Theorem). Let $\Omega \subseteq \mathbb{R}^n$ be open, $f : \Omega \rightarrow \mathbb{R}^n$ be a C^1 function, and $f(\vec{a}) = \vec{b}$. Suppose $Df(\vec{a})$ is invertible (as an $n \times n$ matrix). Then there exists open sets $U, V \subseteq \mathbb{R}^n$ with $\vec{a} \in U$ and $\vec{b} \in V$, and a unique function $g : V \rightarrow U$ with $g(\vec{b}) = \vec{a}$ such that

$$\begin{aligned} g \circ f(\vec{y}) &= \vec{y}, & \forall \vec{y} \in U, \\ f \circ g(\vec{x}) &= \vec{x}, & \forall \vec{x} \in V. \end{aligned}$$

That is g is a local inverse to f . Moreover, g is also a C^1 function with

$$Dg(\vec{x}) = Df(g(\vec{x}))^{-1}, \quad \forall \vec{x} \in V.$$

We note that the Inverse Function Theorem follows from the Implicit Function Theorem:

That is, define $F : f(\Omega) \times \Omega \subseteq \mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{R}^n$ by $F(\vec{x}, \vec{y}) = f(\vec{y}) - \vec{x}$. Then $F(\vec{b}, \vec{a}) = 0$ and $\frac{\partial F}{\partial \vec{y}}|_{(\vec{a}, \vec{b})} = Df(\vec{a})$ is invertible. So by the Implicit Function Theorem, $\vec{y} = g(\vec{x})$ locally with

$$0 = F(\vec{x}, g(\vec{x})) = f \circ g(\vec{x}) - \vec{x}.$$

More precisely, $\vec{y} = g(\vec{x})$ is precisely the local inverse of f as claimed in the Inverse Function Theorem.

However, the Inverse Function Theorem is not just a special case of the Implicit Function Theorem, the two theorems are actually equivalent.

Indeed, suppose the Inverse Function Theorem holds and let $F : \Omega \subseteq \mathbb{R}^{n+k} \rightarrow \mathbb{R}^k$ be a C^1 function with $\frac{\partial F}{\partial \vec{y}}|_{(\vec{a}, \vec{b})}$ be invertible with $F(\vec{a}, \vec{b}) = \vec{c}$. Define $G : \Omega \subseteq \mathbb{R}^{n+k} \rightarrow \mathbb{R}^{n+k}$ by $G(\vec{x}, \vec{y}) = (\vec{x}, F(\vec{x}, \vec{y}))$. Then

$$DG(\vec{a}, \vec{b}) = \begin{pmatrix} I_n & 0 \\ \frac{\partial F}{\partial \vec{x}}(\vec{a}, \vec{b}) & \frac{\partial F}{\partial \vec{y}}(\vec{a}, \vec{b}) \end{pmatrix}.$$

Since $\det(DG(\vec{a}, \vec{b})) = \det \frac{\partial F}{\partial \vec{y}}(\vec{a}, \vec{b}) \neq 0$, $DG(\vec{a}, \vec{b})$ is invertible. Thus, by the Inverse Function Theorem, there exists a local inverse G^{-1} . More precisely, for $(\vec{x}, \vec{y})$ near $(\vec{a}, \vec{b})$ we have

$$\begin{aligned} (\vec{x}, \vec{y}) &= G \circ G^{-1}(\vec{x}, \vec{y}) \\ &= G(G_1^{-1}(\vec{x}, \vec{y}), G_2^{-1}(\vec{x}, \vec{y})) \\ &= (G_1^{-1}(\vec{x}, \vec{y}), F(G_1^{-1}(\vec{x}, \vec{y}), G_2^{-1}(\vec{x}, \vec{y}))). \end{aligned}$$

In particular $\vec{x} = G_1^{-1}(\vec{x}, \vec{y})$ and $\vec{y} = F(\vec{x}, G_2^{-1}(\vec{x}, \vec{y}))$. Setting $\varphi(\vec{x}) = G_2^{-1}(\vec{x}, \vec{c})$, we see that

$$F(\vec{x}, \varphi(\vec{x})) = F(\vec{x}, G_2^{-1}(\vec{x}, \vec{c})) = \vec{c},$$

and φ is precisely the desired function as claimed in the Implicit Function Theorem.

Remark. For a full proof of the Implicit and Inverse Function Theorems see Math3060.

Example 149. Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be the C^1 function $f(x, y) = (x^2 - y^2, 2xy)$. Since $f(-x, -y) = f(x, y)$, f is not injective and so cannot have a global inverse. However, we have that

$$Df(x, y) = \begin{pmatrix} 2x & -2y \\ 2y & 2x \end{pmatrix},$$

which has determinant $4(x^2 + y^2)$. So $Df(x, y)$ is invertible iff $(x, y) \neq (0, 0)$. Therefore, by the Inverse Function Theorem, f is locally invertible about every point other than the origin.

Suppose $g(u, v)$ is such a local inverse of $f(x, y)$ near the point $(x, y) = (1, -1)$. Then, we find that $g(0, -2) = (1, -1)$ with

$$Dg(0, -2) = Df(1, -1)^{-1} = \frac{1}{4} \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}.$$

In both theorems, we assume a certain Jacobian matrix is invertible. Without this assumption, the theorems are inconclusive, and the existence of a local implicit or inverse function is unknown.